28/9/26

PRACTICAL - 8

Aim:

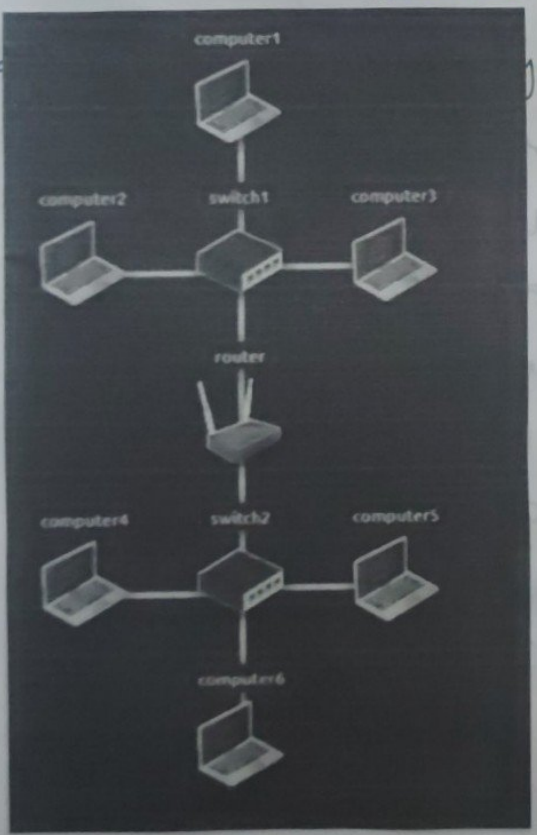
To discover live hosts on a network using Nmap host-discovery scans - ARP scans, ICMP scans and TCP/UDP ping scan - and to understand which protocol operates at which layer of the TCP/IP model, and when each one applies.

1. ARP host discovery:

```
$ sudo nmap -PR -Sn 10.200.6.0/24
```

O/P

```
Starting Nmap 7.80 (https://nmap.org)
Nmap scan report for 10.200.6.1
Host is up
Nmap done: 256 IP address (1 host up)
                scanned in 10.51 seconds
```



ICMP host discovery:

ICMP Echo (-PE)

$ sudo nmap -PE -sn 10.200.6.0/24

Nmap scan report for 10.200.6.1

Host is up (0.069s latency)

Nmap scan report for 10.200.6.50

Host is up (0.070 latency)

Nmap scan report for 10.200.6.250

Host is up (0.069s latency)

Nmap done: 256 IP address (3 hosts up) scanned in 3.38 seconds.

ICMP Timestamp (-PP)

$ nmap -PP -sn 10.200.6.0/24

Nmap scan report for 10.200.6.50

Host is up (0.069s latency)

Nmap scan report for 10.200.6.250

Host is up (0.069s latency)

Nmap done : 256 IP address (3 hosts up) second in 844 second.

ICMP Address Mask (-PM)

$ nMap -PM -sn 10.200.6.0/24

Nmap done : 256 IP address (0 hosts up) scanned in 52.14 seconds.



TCP UDP Ping discovery

i) TCP SYN Ping (-PS)

$ nmap -PS -sn 10.200.6.0/24

Nmap scan report for 10.200.6.50

Host is up (0.07 s latency)

Nmap scan report for 10.200.6.250

ii) TCP ACK Ping (-PA)

Sends a TCP Packet with the ACK flag set

$ sudo nmap -PA -sn 10.200.6.0/24

Nmap scan report for 10.200.6.50

Host is up (0.070 s latency)

Nmap scan report for 10.200.6.250

Host is up (0.069 s latency)

Nmap done: 256 IP address (2 hosts up)
scanned in 36.64 seconds.

iii) UDP Ping

$ sudo nmap -PU -sn 10.200.6.0/24

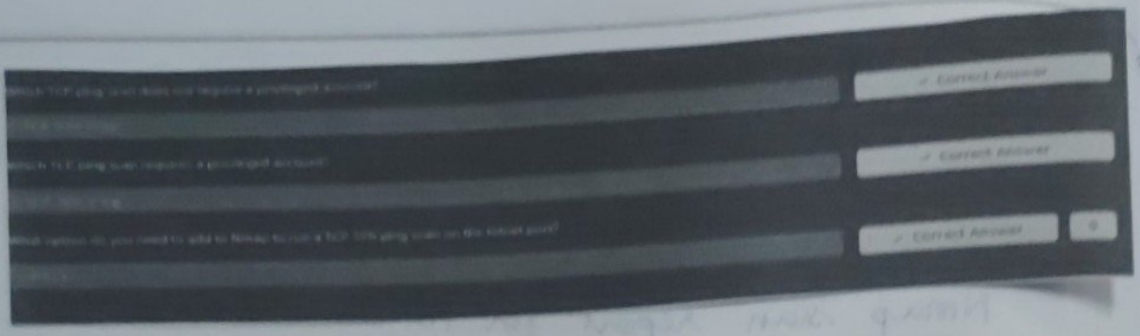
Nmap scan report for 10.200.6.50

Host is up (0.070 latency)

Nmap scan report for 10.200.6.250

Host is up (0.069 latency)

Nmap done: 256 IP address (2 hosts up)
scanned in 85.97 seconds.



Result:

Thus live host discovery for APP, ICMP, TCP UDP was studied and the output is verified successfully.